

SOP-04

Network Health Check Procedure

Department: NOC Operations & Monitoring Team | Safaricom Ethiopia PLC | Version 2.1 | 2026

1. Purpose & Scope

Defines the systematic health verification process conducted at each shift handover to ensure full network visibility and baseline performance compliance.

2. Check Frequency

- â€¢ Mandatory: At start of every shift (06:00, 14:00, 22:00 EAT).
- â€¢ Optional spot checks: every 2 hours during active P1/P2 incidents.

3. RAN Health Checks

- â€¢ Verify Cell Availability Rate (CAR) > 99.5% per region.
- â€¢ Check Call Setup Success Rate (CSSR) > 98.5% nationwide.
- â€¢ Confirm Drop Call Rate (DCR) < 0.5% on voice network.
- â€¢ Review 4G/5G data throughput against baseline (target: >90 Gbps).

4. Transmission Health Checks

- â€¢ Verify all microwave hop link utilization < 70% capacity.
- â€¢ Confirm fiber route redundancy: no single points of failure.
- â€¢ Check IP/MPLS routing table stability & no unexpected convergence.

5. Core Network Checks

- â€¢ Verify MSC/MGW call processing load < 60% CPU utilization.
- â€¢ Confirm SGSN/GGSN/P-GW PDP context activation success rate > 99%.
- â€¢ Check HLR/HSS subscriber data replication is synchronized.

6. Power Systems Checks

- â€¢ Confirm commercial power availability at all Category A sites.
- â€¢ Verify generator fuel levels > 50% at all remote sites.
- â€¢ Check UPS battery health & replace any battery bank < 80% capacity.

7. Reporting

- â€¢ Complete health check checklist in NOC portal within 30 min of shift start.
- â€¢ Escalate any metric below threshold to L2 domain engineer immediately.
- â€¢ Include health check summary in shift handover documentation.